

Asset Pricing HW 1

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1 Q1

1.1 Q1(a)

Starting from the definition of the gross equity return,

$$R_{e,t+1} = \frac{P_{t+1} + D_{t+1}}{P_t},$$

taking logs gives

$$\begin{aligned} r_{e,t+1} &= \log(P_{t+1} + D_{t+1}) - \log(P_t) \\ &= \log \left[P_{t+1} \left(1 + \frac{D_{t+1}}{P_{t+1}} \right) \right] - \log(P_t) \\ &= p_{t+1} + \log(1 + e^{d_{t+1} - p_{t+1}}) - p_t, \end{aligned}$$

where

$$r_{e,t+1} \equiv \log(R_{e,t+1}), \quad p_t \equiv \log(P_t), \quad d_t \equiv \log(D_t).$$

Define

$$f(x) \equiv \log(1 + e^x), \quad x = d_{t+1} - p_{t+1} = dp_{t+1}.$$

We take a first-order Taylor expansion around the fixed average value

$$\bar{x} \equiv \overline{dp}.$$

Thus,

$$f(x) \approx f(\bar{x}) + f'(\bar{x})(x - \bar{x}).$$

Because

$$f'(x) = \frac{e^x}{1 + e^x},$$

define

$$\kappa \equiv \frac{1}{1 + e^{\bar{x}}},$$

so that

$$f'(\bar{x}) = \frac{e^{\bar{x}}}{1 + e^{\bar{x}}} = 1 - \kappa.$$

Therefore,

$$\begin{aligned} f(x) &\approx f(\bar{x}) + (1 - \kappa)(x - \bar{x}) \\ &= \underbrace{f(\bar{x}) - (1 - \kappa)\bar{x}}_{\kappa_0} + (1 - \kappa)x, \end{aligned}$$

where

$$\begin{aligned} \kappa_0 &\equiv \log(1 + e^{\bar{x}}) - (1 - \kappa)\bar{x} \\ &= -\log(\kappa) - (1 - \kappa) \log\left(\frac{1}{\kappa} - 1\right). \end{aligned}$$

Substituting the approximation into the log-return equation gives

$$\begin{aligned} r_{e,t+1} &\approx p_{t+1} + \kappa_0 + (1 - \kappa)(d_{t+1} - p_{t+1}) - p_t \\ &= \kappa_0 + (1 - \kappa)d_{t+1} + \kappa p_{t+1} - p_t. \end{aligned}$$

Using

$$dp_t = d_t - p_t \quad \text{and} \quad \Delta d_{t+1} = d_{t+1} - d_t,$$

we obtain the one-period relation

$$dp_t \approx -\kappa_0 + r_{e,t+1} - \Delta d_{t+1} + \kappa dp_{t+1}.$$

Applying the same relation one period ahead,

$$dp_{t+1} \approx -\kappa_0 + r_{e,t+2} - \Delta d_{t+2} + \kappa dp_{t+2},$$

and substituting this expression into the previous equation gives

$$\begin{aligned} dp_t &\approx -\kappa_0 + r_{e,t+1} - \Delta d_{t+1} \\ &\quad + \kappa(-\kappa_0 + r_{e,t+2} - \Delta d_{t+2} + \kappa dp_{t+2}) \\ &= -\kappa_0(1 + \kappa) + r_{e,t+1} + \kappa r_{e,t+2} \\ &\quad - \Delta d_{t+1} - \kappa \Delta d_{t+2} + \kappa^2 dp_{t+2}. \end{aligned}$$

Continuing this recursive substitution until $t + H$ gives

$$\begin{aligned} dp_t &\approx -\kappa_0(1 + \kappa + \kappa^2 + \dots + \kappa^{H-1}) \\ &\quad + \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} + \kappa^H dp_{t+H}. \end{aligned}$$

For the constant term, define

$$S = 1 + \kappa + \kappa^2 + \dots + \kappa^{H-1}.$$

Multiplying both sides by κ gives

$$\kappa S = \kappa + \kappa^2 + \dots + \kappa^H.$$

Subtracting the second equation from the first,

$$S - \kappa S = 1 - \kappa^H.$$

Therefore,

$$(1 - \kappa)S = 1 - \kappa^H,$$

so that

$$S = \frac{1 - \kappa^H}{1 - \kappa}.$$

Hence,

$$-\kappa_0 (1 + \kappa + \dots + \kappa^{H-1}) = -\kappa_0 \frac{1 - \kappa^H}{1 - \kappa} = \frac{\kappa_0(\kappa^H - 1)}{1 - \kappa}.$$

Thus,

$$dp_t \approx \frac{\kappa_0(\kappa^H - 1)}{1 - \kappa} + \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} + \kappa^H dp_{t+H}. \quad (1.1)$$

Taking conditional expectations at time t , and using

$$E_t[dp_t] = dp_t,$$

gives

$$dp_t \approx \frac{\kappa_0(\kappa^H - 1)}{1 - \kappa} + \sum_{h=1}^H \kappa^{h-1} E_t[r_{e,t+h}] - \sum_{h=1}^H \kappa^{h-1} E_t[\Delta d_{t+h}] + \kappa^H E_t[dp_{t+H}]. \quad (1.2)$$

Since

$$\kappa = \frac{1}{1 + e^{\bar{x}}},$$

we have

$$0 < \kappa < 1,$$

and therefore

$$\kappa^H \rightarrow 0 \quad \text{as} \quad H \rightarrow \infty.$$

To obtain the infinite-horizon relation, impose the no-bubble (or transversality) condition

$$\lim_{H \rightarrow \infty} \kappa^H E_t[dp_{t+H}] = 0.$$

Taking $H \rightarrow \infty$ in Equation (1.2),

$$\frac{\kappa_0(\kappa^H - 1)}{1 - \kappa} \rightarrow -\frac{\kappa_0}{1 - \kappa}.$$

Therefore,

$$dp_t \approx -\frac{\kappa_0}{1 - \kappa} + \sum_{h=1}^{\infty} \kappa^{h-1} E_t[r_{e,t+h}] - \sum_{h=1}^{\infty} \kappa^{h-1} E_t[\Delta d_{t+h}]. \quad (1.3)$$

The derivation requires $P_t > 0$ and $D_t > 0$ so that log prices and log dividends are well defined. It also relies on a first-order Taylor approximation around the fixed average log dividend-price ratio. Finally, the infinite-horizon expression requires the no-bubble (or transversality) condition

$$\lim_{H \rightarrow \infty} \kappa^H E_t[dp_{t+H}] = 0.$$

1.2 Q1(b)

Figure 1 shows that at short horizons, most of the variation in the dividend-price ratio is accounted for by the terminal dividend-price ratio component, while the future return and dividend-growth components are relatively small. As H increases, $b_{dp}^{(H)}$ decreases and approaches zero, showing that the terminal dividend-price ratio becomes less important at longer horizons. At the same time, $b_{re}^{(H)}$ increases steadily and becomes the largest component. The dividend-growth component remains positive but is much smaller than the return component at long horizons. Overall, the results suggest that variation in the aggregate dividend-price ratio is mainly associated with variation in expected future returns rather than expected dividend growth.

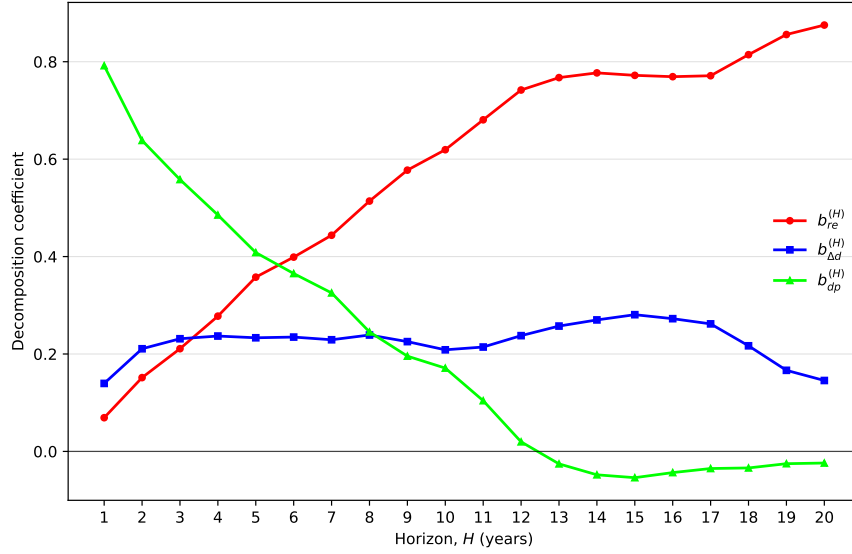


Figure 1: Decomposition of Variation in the Dividend-Price Ratio

1.3 Q1(c)

Figure 2 shows that at short horizons, most of the variation in dp_t is explained by the terminal dividend-price ratio component. As H increases, $b_{dp,VAR}^{(H)}$ declines steadily toward zero, while both the expected-return and expected-dividend-growth components increase. At longer horizons, the two components become similar in magnitude, with the dividend-growth component slightly larger than the return component. This suggests that, under the VAR, long-run variation in the dividend-price ratio is explained by both expected future returns and expected dividend growth, rather than mainly by expected returns alone. Compared with Question 1(b), the VAR-implied decomposition assigns a larger role to expected dividend growth at long horizons.

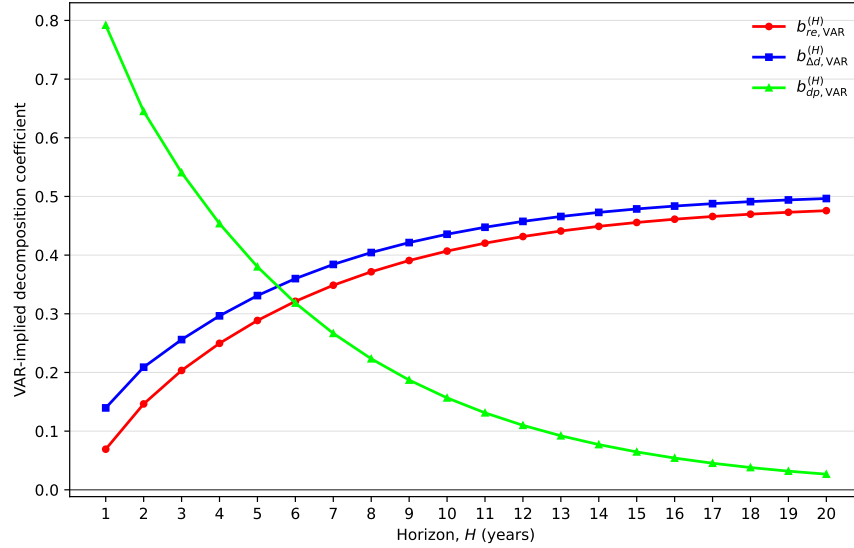


Figure 2: VAR-Implied Decomposition of Variation in the Dividend-Price Ratio

1.4 Q1(d)

The VAR implies

$$b_{re}^{(\infty)} = 0.4899, \quad b_{\Delta d}^{(\infty)} = 0.5089.$$

The results suggest that at the infinite horizon, expected future returns and expected dividend growth contribute almost equally to variation in the dividend-price ratio.

1.5 Q1(e)

By definition,

$$dy_t = \log \left(1 + \frac{D_t}{P_t} \right).$$

Starting from

$$R_{e,t+1} = \frac{P_{t+1} + D_{t+1}}{P_t},$$

taking logs gives

$$r_{e,t+1} = p_{t+1} + \log \left(1 + \frac{D_{t+1}}{P_{t+1}} \right) - p_t = p_{t+1} + dy_{t+1} - p_t.$$

Subtracting

$$\Delta d_{t+1} = d_{t+1} - d_t$$

from both sides yields

$$r_{e,t+1} - \Delta d_{t+1} = p_{t+1} - d_{t+1} + dy_{t+1} + d_t - p_t.$$

Let

$$dp_t = d_t - p_t = \log \left(\frac{D_t}{P_t} \right).$$

Because

$$dy_t = \log(1 + e^{dp_t}),$$

we have

$$e^{dy_t} = 1 + e^{dp_t},$$

and therefore

$$\begin{aligned} dp_t &= \log(e^{dy_t} - 1) \\ &= \log[e^{dy_t}(1 - e^{-dy_t})] \\ &= dy_t + \log(1 - e^{-dy_t}). \end{aligned}$$

Substituting this expression into the preceding identity gives

$$r_{e,t+1} - \Delta d_{t+1} = dy_t + \log(1 - e^{-dy_t}) - \log(1 - e^{-dy_{t+1}}).$$

Now define

$$f(y) = \log(1 - e^{-y}).$$

A first-order Taylor expansion around the average value $\overline{dy}$ gives

$$f(y) \approx f(\overline{dy}) + f'(\overline{dy})(y - \overline{dy}).$$

Since

$$f'(y) = \frac{e^{-y}}{1 - e^{-y}},$$

define

$$\kappa = e^{-\overline{dy}}.$$

It follows that

$$f(\overline{dy}) = \log(1 - \kappa) \quad \text{and} \quad f'(\overline{dy}) = \frac{\kappa}{1 - \kappa}.$$

Thus,

$$\log(1 - e^{-dy_t}) \approx \log(1 - \kappa) + \frac{\kappa}{1 - \kappa}(dy_t - \overline{dy}).$$

Substituting this approximation, we obtain

$$\begin{aligned} r_{e,t+1} - \Delta d_{t+1} &\approx dy_t + \log(1 - \kappa) + \frac{\kappa}{1 - \kappa}(dy_t - \overline{dy}) \\ &\quad - \log(1 - \kappa) - \frac{\kappa}{1 - \kappa}(dy_{t+1} - \overline{dy}) \\ &= \frac{1}{1 - \kappa}dy_t - \frac{\kappa}{1 - \kappa}dy_{t+1}. \end{aligned}$$

Rearranging gives the one-period approximate identity

$$dy_t \approx (1 - \kappa)(r_{e,t+1} - \Delta d_{t+1}) + \kappa dy_{t+1}.$$

Recursively substituting for future values of dy through time $t + H$ gives

$$\begin{aligned} dy_t &\approx (1 - \kappa) \sum_{h=1}^H \kappa^{h-1} (r_{e,t+h} - \Delta d_{t+h}) + \kappa^H dy_{t+H} \\ &= (1 - \kappa) \left(\sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} \right) + \kappa^H dy_{t+H}. \end{aligned} \quad (1.7)$$

Because dy_t is known at time t , taking conditional expectations gives

$$dy_t = (1 - \kappa) \left(\sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t[\Delta d_{t+h}] \right) + \kappa^H \mathbb{E}_t[dy_{t+H}]. \quad (1.8)$$

Finally, impose the no-bubble terminal condition

$$\lim_{H \rightarrow \infty} \kappa^H \mathbb{E}_t[dy_{t+H}] = 0.$$

Taking $H \rightarrow \infty$ in Equation (1.8) therefore yields

$$dy_t = (1 - \kappa) \left(\sum_{h=1}^{\infty} \kappa^{h-1} \mathbb{E}_t[r_{e,t+h}] - \sum_{h=1}^{\infty} \kappa^{h-1} \mathbb{E}_t[\Delta d_{t+h}] \right). \quad (1.9)$$

The derivation requires $P_t > 0$ and $D_t > 0$ so that log prices and log dividends are well defined. The derivation uses a first-order Taylor approximation around the average value $\overline{dy}$ and the no-bubble condition for the infinite-horizon expression.

2 Q2

2.1 Q2(a)

For each horizon $H = 1, \dots, 15$, estimate

$$\frac{1}{H} \sum_{h=1}^H x R_{e,t+h} = a^{(H)} + b^{(H)} \frac{D_t}{P_t} + \varepsilon_t^{(H)},$$

Figure 3 plots the adjusted R^2 across horizons.

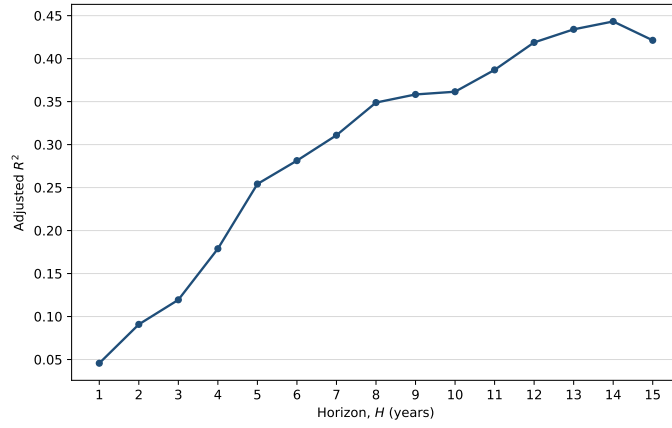


Figure 3: Adjusted R^2 of return-predictive regressions across horizons

Table 1: Predictive regressions of average future excess equity returns

H	$a^{(H)}$	$b^{(H)}$	Adjusted R^2	N
1	-0.0314	2.8038	0.0457	1117
2	-0.0298	2.6858	0.0909	1105
3	-0.0181	2.3903	0.1194	1093
4	-0.0182	2.4173	0.1789	1081
5	-0.0158	2.4006	0.2542	1069
6	-0.0042	2.1229	0.2813	1057
7	0.0016	1.9776	0.3109	1045
8	0.0020	1.9597	0.3489	1033
9	0.0030	1.9109	0.3583	1021
10	0.0051	1.8381	0.3615	1009
11	0.0035	1.8618	0.3869	997
12	0.0019	1.8823	0.4188	985
13	0.0035	1.8406	0.4340	973
14	0.0065	1.7710	0.4432	961
15	0.0120	1.6447	0.4213	949

Note: For each horizon H , the dependent variable is the average of annual simple excess returns at years $t + 1$ through $t + H$. The predictor is D_t/P_t .

The adjusted R^2 generally increases with the return horizon. This suggests that the dividend-price ratio has stronger predictive power for long-horizon excess returns than for short-horizon excess returns.

2.2 Q2(b)

Estimate

$$xR_{e,t+1} = a + b \frac{D_t}{P_t} + \varepsilon_t.$$

The same coefficient estimate is obtained under all five methods, but the estimated standard errors and t -statistics differ.

Table 2: Predictive-regression inference for excess equity returns

Method	b	Standard error	t -statistic
OLS	2.8038	0.3802	7.38
White	2.8038	0.6587	4.26
Newey-West (11)	2.8038	1.2990	2.16
Hansen-Hodrick (11)	2.8038	1.4537	1.93
Newey-West (data-driven)	2.8038	1.2523	2.24

Note: All rows use the same OLS slope estimate and $N = 1117$ observations. Newey-West (11) uses Bartlett weights; Hansen-Hodrick (11) uses uniform weights. The data-driven Newey-West bandwidth is $L = 22$. It is obtained without prewhitening: the preliminary lag is $q = 6$, the estimated plug-in constant is 2.1278, and the unrounded bandwidth is 22.0776, which is rounded down to $L = 22$.

The OLS standard error is the smallest, giving the largest t -statistic. The White standard error is larger because it allows for heteroskedasticity, but it still does not account for autocorrelation. The Newey-West and Hansen-Hodrick standard errors are larger because they account for the autocorrelation created by overlapping annual returns in the monthly data. As a result, the corresponding t -statistics are much lower.

Using the Newey-West (1994) automatic bandwidth procedure, the preliminary lag is $q = \lfloor 4(1117/100)^{2/9} \rfloor = 6$. The estimated quantities are $S_0 = 0.0009458904$ and $S_1 = 0.0023971961$, which imply the plug-in constant $\widehat{\text{const}} = 1.1447 \left[(S_1/S_0)^2 \right]^{1/3} = 2.1278$. Therefore, the data-driven bandwidth is $L = \left\lfloor \widehat{\text{const}} \times 1117^{1/3} \right\rfloor = 22$. Using this bandwidth, the estimated standard error is 1.2523, with a corresponding t -statistic of 2.24.

Overall, the results show that inference is sensitive to the treatment of serial correlation. Ignoring autocorrelation makes the predictive relation appear much more statistically significant.

2.3 Q2(c)

Following Amihud and Hurvich (2004), first, estimate

$$\frac{D_{t+1}}{P_{t+1}} = \theta + \phi \frac{D_t}{P_t} + u_{t+1},$$

which gives

$$\hat{\theta} = 0.0111, \quad \hat{\phi} = 0.7197.$$

Therefore, the predictor $\frac{D_t}{P_t}$ is consistent.

Using $T = 94$, the bias-corrected persistence coefficient is

$$\hat{\phi}^c = \hat{\phi} + \frac{1}{T}(1 + 3\hat{\phi}) + \frac{3}{T^2}(1 + 3\hat{\phi}) = 0.7544.$$

Then construct

$$\hat{u}_{t+1}^c = \frac{D_{t+1}}{P_{t+1}} - \left(\hat{\theta} + \hat{\phi}^c \frac{D_t}{P_t} \right),$$

and estimate

$$xR_{e,t+1} = a + b \frac{D_t}{P_t} + b_u \hat{u}_{t+1}^c + \varepsilon_t.$$

The estimates are

$$\hat{b}_{2c} = 2.3362, \quad \hat{b}_u = -13.4837.$$

For comparison,

$$\hat{b}_{2b} = 2.8038, \quad \hat{b}_{2c} - \hat{b}_{2b} = -0.4676.$$

The bias correction reduces the estimated predictive coefficient, consistent with the finite-sample bias induced by a persistent predictor.

2.4 Q2(d)

The predictive regression is evaluated using an expanding-window out-of-sample procedure. For each forecast date $t + 1$, only information available through time t is used to estimate

$$xR_{e,t+1} = a_t + b_t \frac{D_t}{P_t} + \varepsilon_t.$$

The corresponding out-of-sample forecast is

$$\hat{E}_t^{OS}[xR_e] = \hat{a}_t + \hat{b}_t \frac{D_t}{P_t}.$$

The benchmark forecast, $\bar{x}R_{e,t}$, is the expanding-window average of realized excess returns over the same historical sample. The in-sample fitted value is

$$\hat{E}_t^{IS}[xR_e] = \hat{a}^{IS} + \hat{b}^{IS} \frac{D_t}{P_t},$$

where $\hat{a}^{IS}$ and $\hat{b}^{IS}$ are estimated using the full sample.

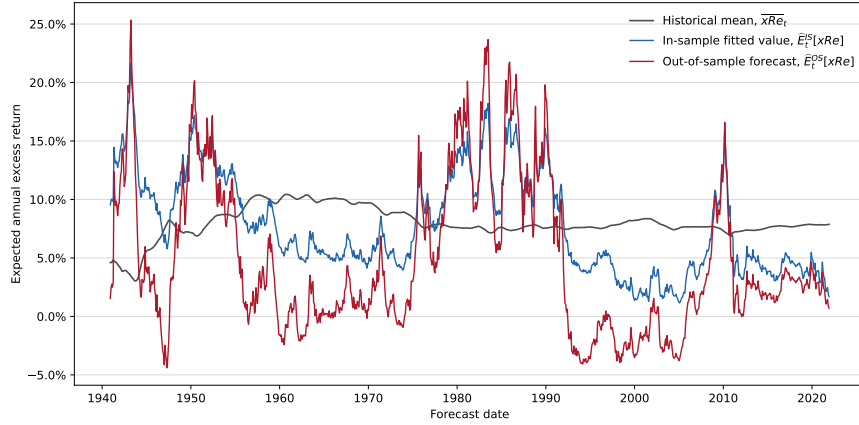


Figure 4: Historical mean, in-sample fitted values, and out-of-sample forecasts

Figure 4 shows that the out-of-sample forecast varies substantially over time and is generally more volatile than the historical mean. The in-sample fitted value and the out-of-sample forecast often move in the same direction, although they differ considerably in some periods.

The overall out-of-sample R^2 is

$$R_{OS}^2 = 1 - \frac{\sum_t \left(xR_{e,t+1} - \hat{E}_t^{OS}[xR_e] \right)^2}{\sum_t \left(xR_{e,t+1} - \bar{x}R_{e,t} \right)^2} = -0.0061.$$

The out-of-sample period runs from December 1940 to December 2021 and contains 973 forecasts. Since R_{OS}^2 is slightly negative, the dividend-price ratio

model does not outperform the expanding-window historical-mean benchmark over the full out-of-sample period.

To examine changes in forecasting performance over time, R_{OS}^2 is also calculated using a 50-year rolling window of forecast errors, while the estimation of a_t and b_t remains expanding-window.

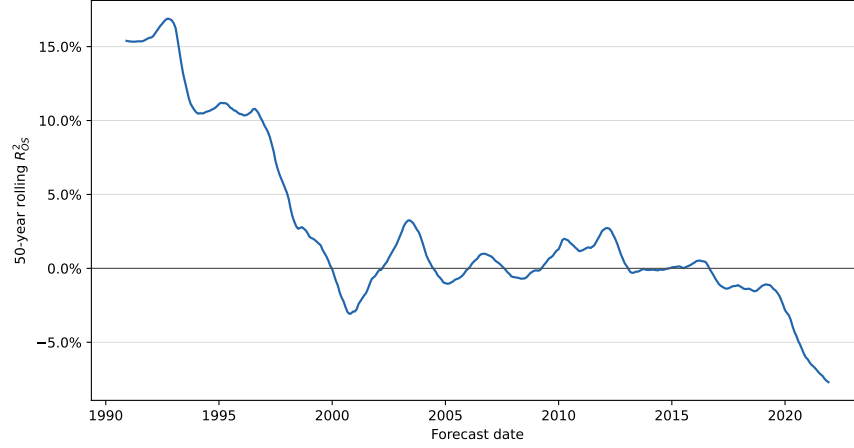


Figure 5: 50-year rolling out-of-sample R_{OS}^2

Figure 5 shows that the rolling R_{OS}^2 is strongly positive in the early part of the sample but declines over time. It becomes negative around 2000, returns close to zero or slightly positive in some later periods, and becomes clearly negative again near the end of the sample. The out-of-sample predictive performance of the dividend-price ratio is therefore not stable over time.

2.5 Q2(e)

The out-of-sample exercise in Q2(d) is repeated under the steady-state restrictions

$$\hat{a}_t = G_t - 1, \quad \hat{b}_t = G_t,$$

where G_t is the expanding-window historical average of $e^{\Delta d_t}$ computed over the same historical sample used for the benchmark forecast.

The restricted out-of-sample forecast is

$$\hat{E}_t^{OS}[xR_e] = (G_t - 1) + G_t \frac{D_t}{P_t}.$$

The same expanding-window historical mean, $\bar{xR}_{e,t}$, and the same in-sample fitted value,

$$\hat{E}_t^{IS}[xR_e] = \hat{a}^{IS} + \hat{b}^{IS} \frac{D_t}{P_t},$$

are used for comparison.

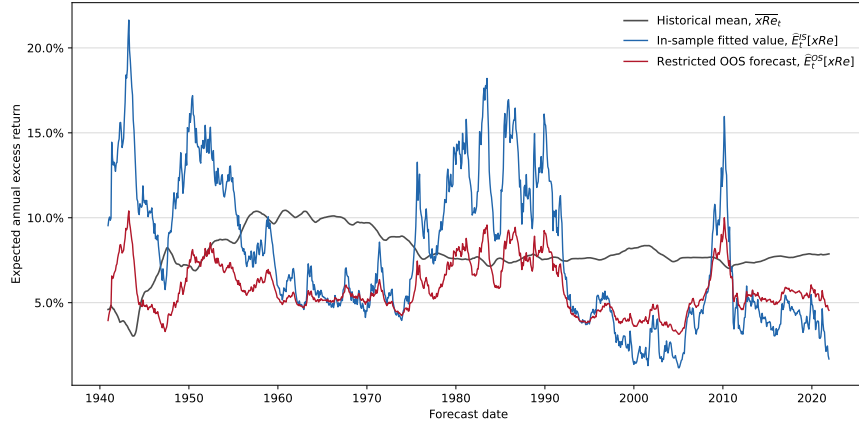


Figure 6: Historical mean, in-sample fitted values, and restricted out-of-sample forecasts

The overall out-of-sample R^2 is

$$R_{OS}^2 = 1 - \frac{\sum_t \left(xR_{e,t+1} - \hat{E}_t^{OS}[xR_e] \right)^2}{\sum_t \left(xR_{e,t+1} - \bar{xR}_{e,t} \right)^2} = 0.0321.$$

The out-of-sample period contains 973 forecasts from December 1940 to December 2021. Since R_{OS}^2 is positive, the restricted model outperforms the expanding-window historical-mean benchmark over the full out-of-sample period.

A 50-year rolling R_{OS}^2 is also calculated using the same expanding-window estimation procedure.

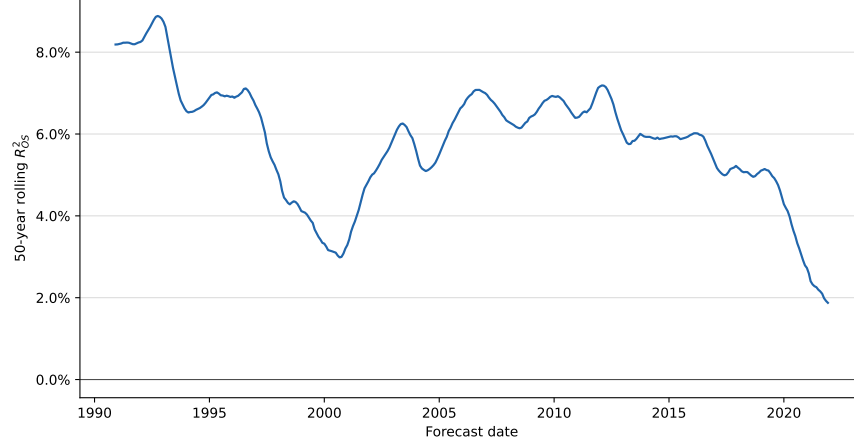


Figure 7: 50-year rolling out-of-sample R_{OS}^2 under the steady-state restrictions

The rolling R_{OS}^2 remains positive throughout the sample. It is about 0.082 at the beginning of the rolling period and declines to about 0.019 by the end of the sample. Compared with Q2(d), the steady-state restrictions improve out-of-sample forecasting performance.

3 Q3

3.1 Q3(a)

The CRSP monthly stock data are downloaded from CRSP and cover the period from January 1962 to December 2025. Earlier observations are used only to construct the momentum signal. The comparison with the Chen and Zimmermann momentum measure and the monthly cross-sectional regressions begin in June 1963.

For each stock and month τ , construct the momentum signal as the cumulative return from month $\tau - 12$ to month $\tau - 1$,

$$MOM_{i,\tau} = \prod_{h=1}^{12} (1 + R_{i,\tau-h}) - 1.$$

Then merge the constructed signal with the Chen and Zimmermann momentum measure, $MOM_{i,\tau}^{CZ}$, and estimate the following cross-sectional regression separately in each month:

$$MOM_{i,\tau}^{CZ} = a_{\tau} + b_{\tau} MOM_{i,\tau} + \varepsilon_{i,\tau}.$$

Figure 8 plots the monthly intercept estimates.

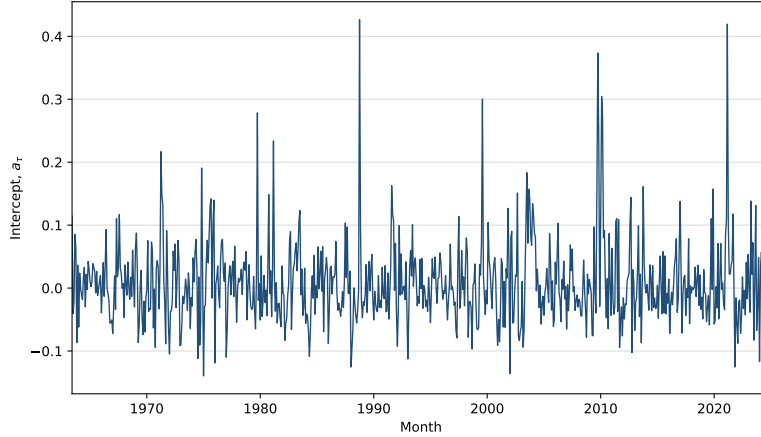


Figure 8: Monthly cross-sectional regression intercepts

Figure 9 plots the monthly slope estimates.

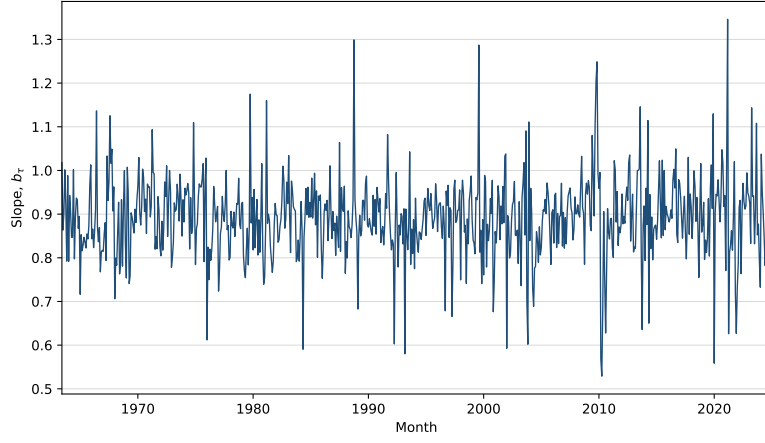


Figure 9: Monthly cross-sectional regression slopes

Figure 10 plots the monthly R^2 values.

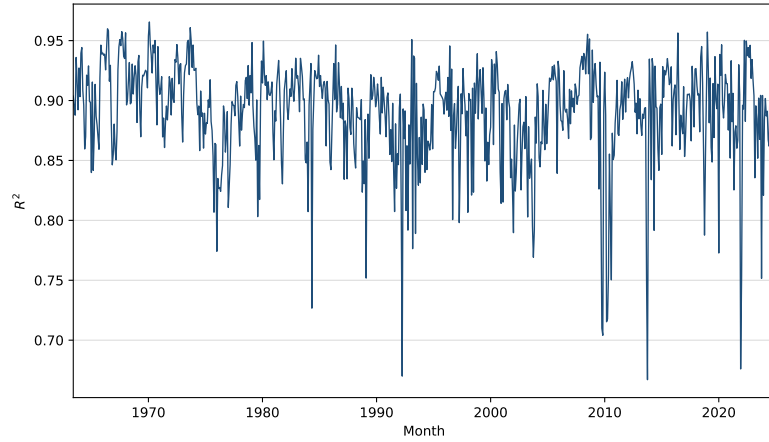


Figure 10: Monthly cross-sectional regression R^2

The intercepts are generally close to zero, while the slopes are mostly around 0.8 to 1.0. The R^2 values are also high in most months, generally around 0.85 to 0.95. Overall, the constructed momentum signal is strongly related to the Chen and Zimmermann measure.

3.2 Q3(b)

Book-to-market is constructed using book equity from Compustat and market equity from CRSP. Book equity is calculated as

$$BE = SE + TXDITC - BVPS,$$

and market equity is calculated as

$$ME = |PRC| \times SHROUT.$$

For June of year t , BM uses accounting information from year $t - 1$ and market equity from December of year $t - 1$. The resulting BM is kept fixed from June of year t through May of year $t + 1$.

For each month τ , estimate the cross-firm regression

$$BM_{j,\tau}^{CZ} = a_{\tau} + b_{\tau} BM_{j,\tau} + \varepsilon_{j,\tau},$$

using

$$BM^{CZ} = BM_{dec}.$$

The downloaded Chen and Zimmermann (2022) data show that BM_{dec} is already a book-to-market ratio rather than its logarithm, so it is used directly rather than exponentiated.

Figure 11 plots the monthly intercept estimates.

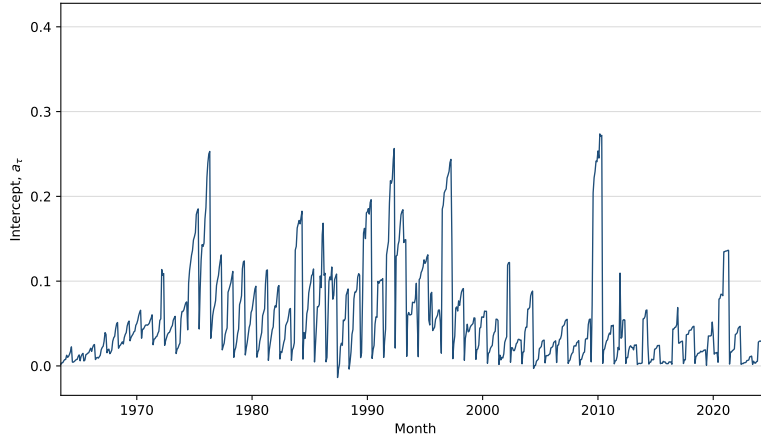


Figure 11: Monthly cross-sectional regression intercepts

Figure 12 plots the monthly slope estimates.

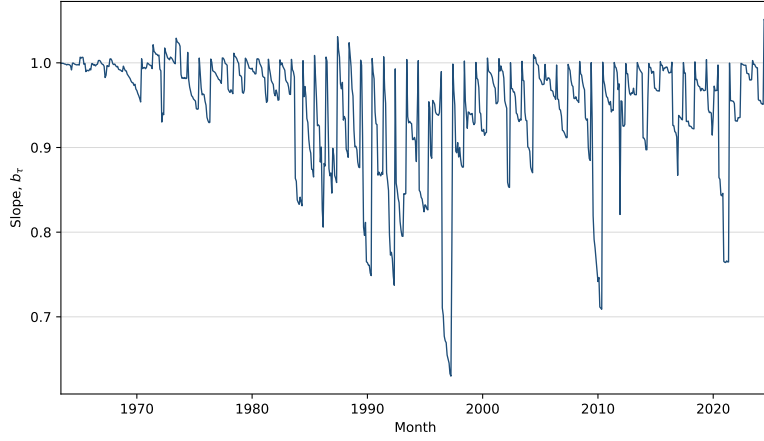


Figure 12: Monthly cross-sectional regression slopes

Figure 13 plots the monthly R^2 values.

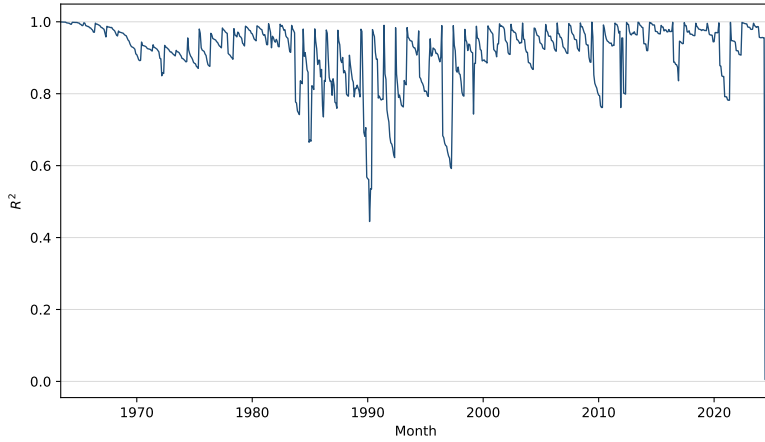


Figure 13: Monthly cross-sectional regression R^2

The slopes are generally close to one and the R^2 values are generally high, indicating that the constructed BM measure closely matches the Chen and Zimmermann measure. The intercepts are also generally small, although there are some periods with larger differences.

The sharp drop in R^2 at the end of the sample is caused by one extreme firm rather than a general deterioration in the match between the two BM measures. From June to November 2024, PERMNO 18558 has

$$BM = 1.009378, \quad BM^{CZ} = 1096.598315.$$

This single observation has an unusually large value of BM^{CZ} while its constructed BM is ordinary, and therefore strongly affects the monthly cross-sectional regressions. The reported R^2 is only about 0.0048–0.0050 from June to November 2024. After excluding this one observation, the corresponding R^2 values increase to about 0.9927–0.9958. In December 2024, the reported R^2 returns to 0.9841. Therefore, the end-of-sample decline is driven by this single extreme observation rather than a broad decline in the accuracy of the constructed BM measure.

3.3 Q3(c)

Using the Chen and Zimmermann signals BM^{CZ} , MOM^{CZ} , and GP^{CZ} , construct decile portfolios under five different portfolio-formation methods. For each method, calculate the average monthly excess return of each decile portfolio. HML is defined as the return of decile 10 minus decile 1.

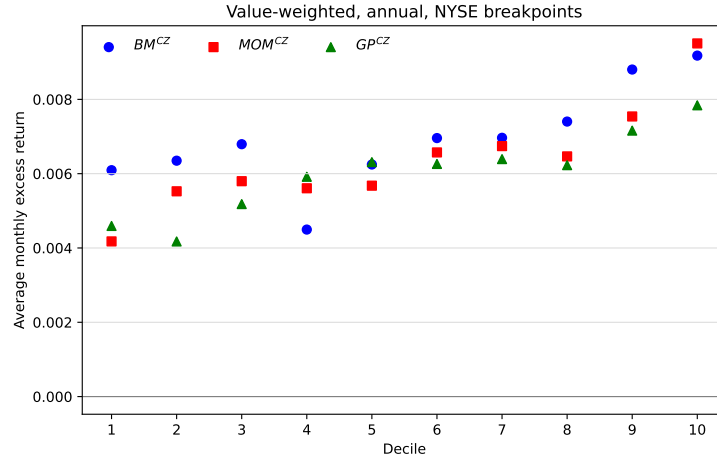


Figure 14: Value-weighted portfolios, annual rebalancing, NYSE breakpoints

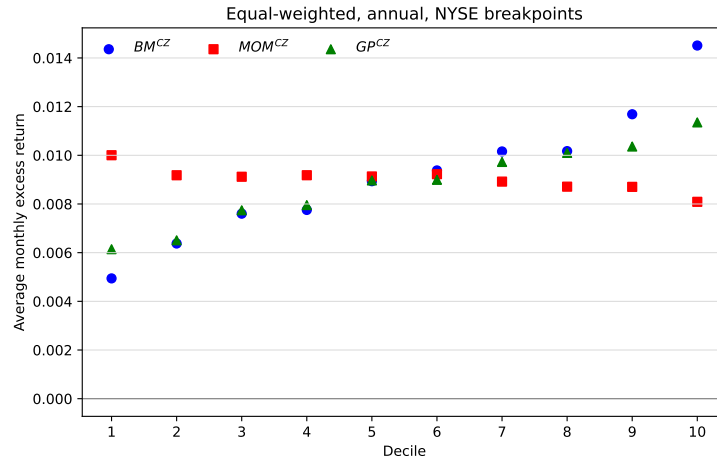


Figure 15: Equal-weighted portfolios, annual rebalancing, NYSE breakpoints

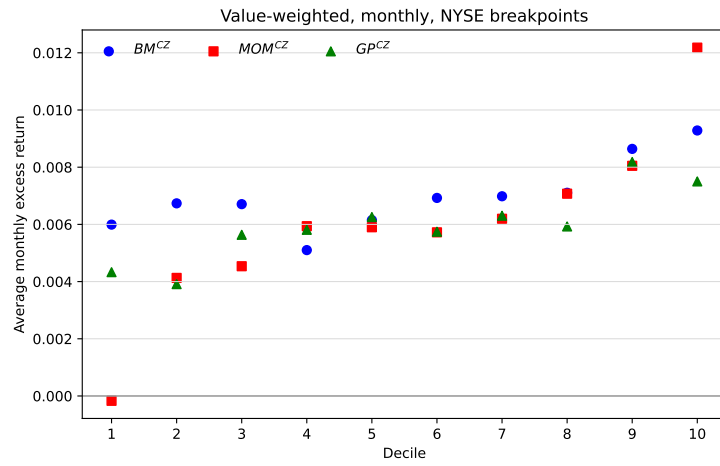


Figure 16: Value-weighted portfolios, monthly rebalancing, NYSE breakpoints

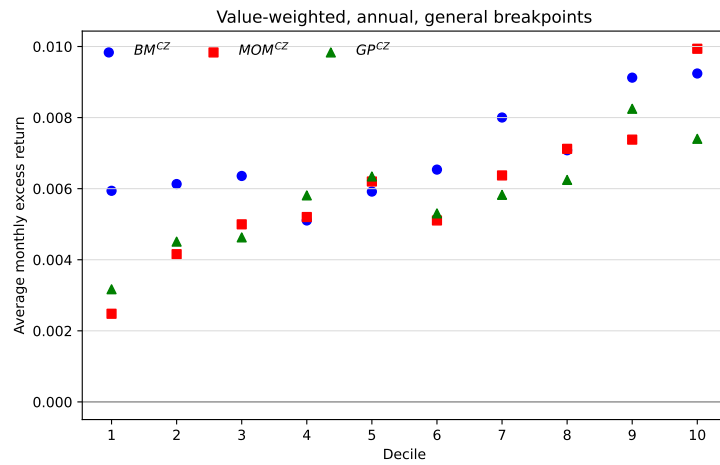


Figure 17: Value-weighted portfolios, annual rebalancing, general breakpoints

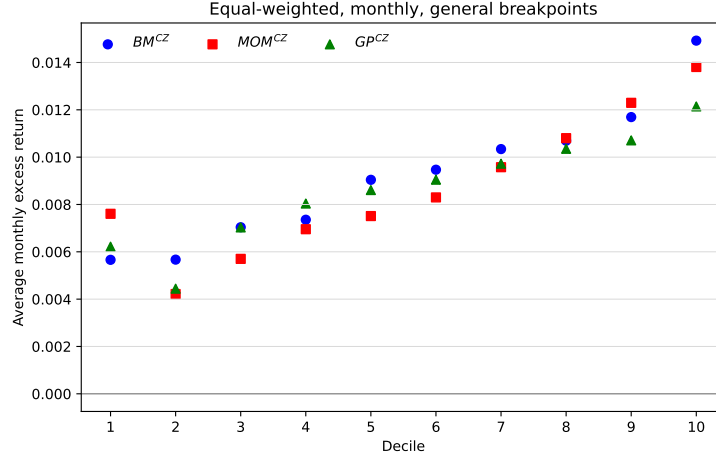


Figure 18: Equal-weighted portfolios, monthly rebalancing, general breakpoints

Table 3: High-minus-low excess returns for Chen–Zimmermann signals

Signal	Portfolio construction	Average HML (%)	t -statistic
BM^{CZ}	Value-weighted, annual, NYSE breakpoints	0.31%	1.47
BM^{CZ}	Equal-weighted, annual, NYSE breakpoints	0.96%	4.93
BM^{CZ}	Value-weighted, monthly, NYSE breakpoints	0.33%	1.62
BM^{CZ}	Value-weighted, annual, general breakpoints	0.33%	1.36
BM^{CZ}	Equal-weighted, monthly, general breakpoints	0.93%	4.76
MOM^{CZ}	Value-weighted, annual, NYSE breakpoints	0.53%	2.36
MOM^{CZ}	Equal-weighted, annual, NYSE breakpoints	-0.19%	-0.89
MOM^{CZ}	Value-weighted, monthly, NYSE breakpoints	1.23%	4.53
MOM^{CZ}	Value-weighted, annual, general breakpoints	0.75%	2.59
MOM^{CZ}	Equal-weighted, monthly, general breakpoints	0.62%	1.96
$GPCZ$	Value-weighted, annual, NYSE breakpoints	0.32%	1.99
$GPCZ$	Equal-weighted, annual, NYSE breakpoints	0.52%	3.20
$GPCZ$	Value-weighted, monthly, NYSE breakpoints	0.32%	1.88
$GPCZ$	Value-weighted, annual, general breakpoints	0.42%	2.05
$GPCZ$	Equal-weighted, monthly, general breakpoints	0.59%	2.88

Note: HML is the monthly excess return of decile 10 minus decile 1. Signals observed in month τ earn returns in month $\tau + 1$. Value-weighted returns use market equity from the previous month. The t -statistics use the Newey–West (1987, 1994) automatic-bandwidth procedure without prewhitening or recoloring. Selected bandwidths range from $L = 1$ to $L = 12$. Average HML returns are reported as monthly percentages.

The portfolio returns generally increase across deciles for BM^{CZ} and $GPCZ$, indicating that stocks with higher book-to-market ratios and higher profitability tend to earn higher average excess returns. The pattern for MOM^{CZ} depends more on the portfolio construction method.

The HML results also vary across weighting and rebalancing methods. For BM^{CZ} , the HML return is much stronger for equal-weighted portfolios than for value-weighted portfolios. For MOM^{CZ} , the strongest HML result is obtained with monthly rebalancing and NYSE breakpoints, while the equal-weighted annual specification gives a negative HML return. The GP^{CZ} HML returns are positive across all five specifications.

3.4 Q3(d)

For each month τ , cross-sectional regressions of next-month stock excess returns are estimated using the quantiles of BM^{CZ} , GP^{CZ} , and Dur . For each signal, the quantile is constructed as its monthly cross-sectional percentile rank using average ranks for ties, so that

$$Q_{j,\tau}^X \in (0, 1].$$

Signals dated τ are used to explain excess returns dated $\tau + 1$. Each of the seven specifications is estimated using both OLS and WLS, where WLS uses month- τ market equity as the weight.

To make the specifications directly comparable, all regressions use the common sample from June 1973 through December 2024, corresponding to next-month returns from July 1973 through January 2025. This gives 619 monthly cross-sectional regressions for each specification. Within each month, each specification uses its own complete-case firm sample.

The Fama–MacBeth coefficient estimates are obtained by averaging the monthly cross-sectional coefficients over time. Coefficient estimates in the table are multiplied by 100 and are therefore reported in monthly percentage points.

The OLS results show that Q^{BM} and Q^{GP} are positively related to future excess returns, while Q^{Dur} is negatively related to future excess returns. When all three signals are included together, the OLS coefficients remain positive for Q^{BM} and Q^{GP} and negative for Q^{Dur} , all statistically significant at the 1% level.

The WLS results are generally weaker. In the full specification, Q^{GP} remains positive and statistically significant, while the coefficients on Q^{BM} and Q^{Dur} are smaller and not statistically significant. Overall, higher book-to-market and profitability are associated with higher future excess returns, while higher equity duration is associated with lower future excess returns. These relations are stronger in the OLS regressions than in the market-equity-weighted regressions.

Table 4: Fama–MacBeth regressions of next-month stock excess returns: conventional inference

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Panel A: OLS</i>							
Intercept	0.5371* (1.88)	0.5366* (1.74)	1.6126*** (7.04)	0.0412 (0.12)	1.3254*** (5.67)	1.4262*** (5.48)	0.6962** (2.51)
Q^{BM}	0.9339*** (5.18)			1.0433*** (5.62)	0.3581* (1.67)		0.6536*** (2.65)
Q^{GP}		0.8145*** (4.88)		0.8593*** (5.06)		0.2595* (1.80)	0.5441*** (3.30)
Q^{Dur}			−1.1370*** (−6.51)		−0.9527*** (−4.94)	−1.0555*** (−5.68)	−0.6005*** (−2.68)
Months	619	619	619	619	619	619	619
<i>Panel B: WLS</i>							
Intercept	0.6059*** (2.85)	0.4230* (1.85)	1.3395*** (6.60)	0.1215 (0.45)	1.3377*** (4.41)	0.7975** (2.39)	0.4591 (1.16)
Q^{BM}	0.2815 (1.17)			0.6263** (2.54)	−0.2078 (−0.64)		0.3042 (1.09)
Q^{GP}		0.4190** (2.22)		0.6347*** (3.50)		0.5594** (2.12)	0.6361** (2.40)
Q^{Dur}			−0.9190*** (−3.82)		−0.8314*** (−2.70)	−0.5585* (−1.79)	−0.2680 (−0.77)
Months	619	619	619	619	619	619	619

Note: Each cell reports the time-series average of the monthly cross-sectional coefficient on the first line, with its ordinary Fama–MacBeth t -statistic in parentheses below. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. Signal quantiles are monthly percentile ranks calculated using average ranks for ties. Signals dated τ explain excess returns dated $\tau + 1$. WLS uses market equity dated τ . Each specification uses its own within-month complete-case firm sample. All specifications use the common signal-month sample from June 1973 through December 2024, with returns from July 1973 through January 2025. Coefficient estimates are reported in monthly percentage points; the underlying signal quantiles remain on their 0-to-1 percentile-rank scale. Each specification contains 619 monthly estimates.

3.5 Q3(e)

Decile portfolios are formed separately using BM^{CZ} , GP^{CZ} , and Dur . The portfolios are rebalanced annually in June using NYSE breakpoints. Both value-weighted and equal-weighted portfolios are considered.

For each portfolio p and month τ , the average signal decile of the firms in the portfolio is calculated. These portfolio-level measures are denoted by $Dec_{p,\tau}^{BM}$, $Dec_{p,\tau}^{GP}$, and $Dec_{p,\tau}^{Dur}$. Signals dated τ are used to explain portfolio excess returns dated $\tau + 1$.

The seven specifications are estimated using pooled OLS. Driscoll–Kraay standard errors are calculated using Bartlett weights and an automatic bandwidth rule without prewhitening. The common signal-month sample runs from June 1973 through December 2024, with returns from July 1973 through January 2025.

For the value-weighted portfolios, the univariate coefficients on Dec^{BM} and Dec^{GP} are positive but relatively weak, while the coefficient on Dec^{Dur} is negative and statistically significant. In the full specification, the duration coefficient remains negative, while the BM and GP coefficients are close to zero.

For the equal-weighted portfolios, Dec^{BM} and Dec^{GP} have positive and statistically significant coefficients in the univariate regressions, while Dec^{Dur} has a negative and statistically significant coefficient. When all three signals are included together, the coefficients become weaker. Overall, the portfolio regressions provide stronger evidence for a negative relation between equity duration and future returns, while the BM and GP relations are more sensitive to the portfolio weighting scheme and the inclusion of other signals.

Table 5: Portfolio-level multivariate return regressions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Panel A: Value-weighted portfolios</i>							
Intercept	0.5830*** (2.84)	0.4924** (2.12)	1.1840*** (5.77)	0.2425 (0.96)	0.9795*** (3.18)	1.3133*** (4.08)	1.2660** (1.98)
Dec^{BM}	0.0340 (1.43)			0.0444* (1.94)	0.0220 (0.68)		0.0021 (0.05)
Dec^{GP}		0.0305 (1.63)		0.0409** (2.23)		-0.0211 (-0.86)	-0.0161 (-0.48)
Dec^{Dur}			-0.0642*** (-3.32)		-0.0549* (-1.89)	-0.0823*** (-3.22)	-0.0794* (-1.73)
Observations	6190	6190	6190	12380	12380	12380	18570
DK bandwidth	3	2	2	1	2	1	1
<i>Panel B: Equal-weighted portfolios</i>							
Intercept	0.5161* (1.76)	0.6022* (1.90)	1.5487*** (6.31)	-0.2706 (-0.72)	1.2494*** (3.60)	1.6284*** (4.79)	0.9060 (1.29)
Dec^{BM}	0.0910*** (4.57)			0.1231*** (4.96)	0.0436 (1.24)		0.0584 (1.16)
Dec^{GP}		0.0680*** (4.58)		0.1055*** (5.42)		-0.0002 (-0.01)	0.0376 (1.10)
Dec^{Dur}			-0.0948*** (-6.09)		-0.0855*** (-2.81)	-0.1148*** (-5.20)	-0.0798* (-1.75)
Observations	6190	6190	6190	12380	12380	12380	18570
DK bandwidth	3	5	4	5	4	4	5

Note: Pooled OLS estimates. Parentheses contain Driscoll–Kraay t -statistics based on monthly pooled score sums, Bartlett weights, and the no-prewhitening Newey–West (1994) automatic bandwidth rule used in Question 2(b). *, **, and *** denote $|t| \geq 1.645$, $|t| \geq 1.96$, and $|t| \geq 2.576$, respectively. Coefficient estimates are reported in monthly percentage points; the underlying pooled regressions remain in decimal return units. Portfolios are formed each June using NYSE breakpoints. Signals dated τ explain returns dated $\tau + 1$. The common signal-month sample is June 1973 through December 2024; returns run from July 1973 through January 2025.

4 Q4

4.1 Q4(a)

Using the Fama-Bliss bond yield data, log yields, forward rates, and annual bond returns are constructed for maturities from one to five years. Yield spreads $xy^{(H)}$, forward spreads $xf^{(H)}$, and excess bond returns $xr^{(H)}$ are then calculated relative to the one-year bond. Table 6 reports the sample averages for $H = 2, 3, 4, 5$.

Table 6: Average yield, forward-rate, and bond-return spreads

	$H = 2$	$H = 3$	$H = 4$	$H = 5$
Mean $xy(H)$	0.001686	0.003278	0.004660	0.005639
Mean $xf(H)$	0.003372	0.006461	0.008805	0.009557
Mean $xr(H)$	0.003154	0.006069	0.008198	0.008719

The average yield and forward spreads are positive and generally increase with maturity. Average excess bond returns are also positive for all maturities and are larger for longer-maturity bonds.

4.2 Q4(b)

Question 4(b) examines whether the current yield spread predicts future hold-to-maturity excess bond returns. For each maturity H , the average annual excess return from t to $t + H$ is regressed on the current yield spread $xy_t^{(H)}$.

Table 7: Regressions of hold-to-maturity excess returns on yield spreads

	$H = 2$	$H = 3$	$H = 4$	$H = 5$
$b^{(H)}$ [t-statistic]	0.677 [3.57]	0.532 [3.21]	0.414 [2.52]	0.345 [2.30]
R^2	0.077	0.052	0.038	0.032
Observations	859	847	835	823

Note: OLS estimates. Brackets contain Hansen-Hodrick (1980) t-statistics using uniform weights and $L = 12H - 1$ monthly lags.

The estimated slope coefficients are positive for all maturities, indicating that larger yield spreads are associated with higher future hold-to-maturity excess bond returns. The estimated slope declines as maturity increases.

4.3 Q4(c)

Question 4(c) studies one-year-ahead return predictability using forward spreads. For each maturity, the next one-year excess bond return is regressed on the current forward spread $xf_t^{(H)}$.

Table 8: Regressions of one-year excess returns on forward spreads

	$H = 2$	$H = 3$	$H = 4$	$H = 5$
$b^{(H)}$ [t-statistic]	0.677 [3.21]	0.889 [3.33]	1.116 [3.59]	0.964 [2.91]
R^2	0.077	0.086	0.107	0.067
Observations	859	859	859	859

Note: OLS estimates. Brackets contain Bartlett Newey-West (1987) t-statistics using the automatic bandwidth procedure in Newey and West (1994).

The forward-spread coefficients are positive across all maturities. Forward spreads contain information about future one-year excess bond returns.

4.4 Q4(d)

The Cochrane-Piazzesi factor is estimated by regressing the average one-year-ahead excess return across the two- to five-year bonds on the current one-through five-year forward rates. The fitted value from this regression defines the factor cp_t .

Figure 19 plots the time series of the Cochrane-Piazzesi factor, cp_t .

Table 9: Cochrane-Piazzesi factor coefficients

Coefficient	Estimate
$\hat{\theta}_0$	-0.013448
$\hat{\theta}_1$	-0.987387
$\hat{\theta}_2$	-0.211534
$\hat{\theta}_3$	0.809338
$\hat{\theta}_4$	1.084352
$\hat{\theta}_5$	-0.464740

Note: OLS estimates. $R^2 = 0.158$; observations: 859.

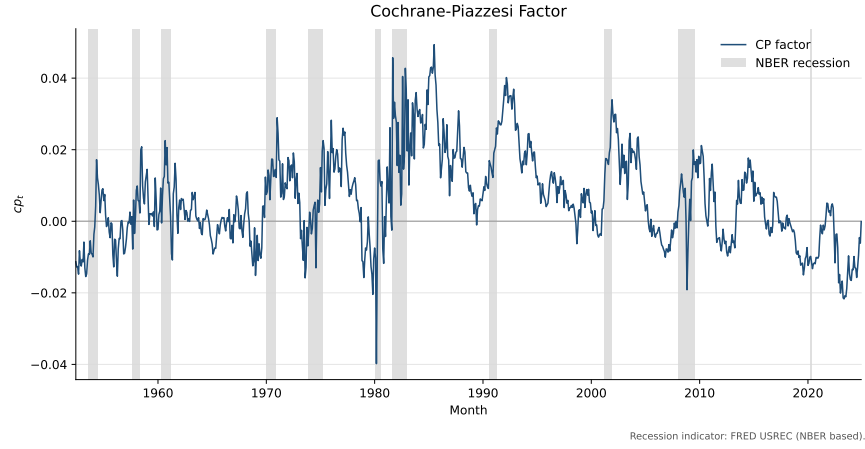


Figure 19: CP factor Time Series

4.5 Q4(e)

Question 4(e) examines whether the Cochrane-Piazzesi factor predicts one-year excess returns for individual maturities. For $H = 2, 3, 4, 5$, the next one-year excess bond return is regressed on cp_t .

Table 10: One-year excess-return predictability using the Cochrane-Piazzesi factor

	$H = 2$	$H = 3$	$H = 4$	$H = 5$
$b^{(H)}$ [t-statistic]	0.442 [4.13]	0.827 [4.12]	1.252 [4.39]	1.479 [4.20]
R^2	0.137	0.144	0.170	0.155
Observations	859	859	859	859

Note: OLS estimates. Brackets contain Bartlett Newey-West (1987) t-statistics using the automatic bandwidth procedure in Newey and West (1994).

The estimated coefficients on cp_t are positive for all maturities and generally increase with maturity. The R^2 values are between 0.137 and 0.170, which are generally larger than those from the single-forward-spread regressions in 4(c).